

Brittle materials-based fully stretchable organic light-emitting diodes

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Stretchable small-molecule organic light-emitting diodes (OLEDs) are hampered by random cracking and efficiency degradation of inherently brittle small-molecule organic semiconductors (SMOS) under mechanical deformation, severely limiting their practical application. Herein, we propose an interfacial stress regulation strategy based on a hierarchical architecture. Departing from the conventional crack-suppression paradigm in stretchable optoelectronics, our strategy employs active crack engineering to guide controlled nanocrack formation, enabling the fabrication of nanocracked, fully stretchable OLEDs compatible with inherently brittle SMOS while achieving both high efficiency and excellent stretchability. The devices achieve a maximum brightness of 16,050 cd/m², current efficiency of 85.7 cd/A, external quantum efficiency of 24.7%, and 250% tensile strain. Stable operation is maintained under complex deformations, while the strategy demonstrates broad universality for multicolor emission. This work offers a general strategy for integrating brittle high-performance SMOS into stretchable displays, advancing the development of wearable and conformal electronics.

Stretchable displays, enabling the seamless integration of visual functionality with dynamic deformability, are poised to revolutionize next-generation wearable/implantable electronics by breaking the long-standing paradigm constraints of rigid/flexible in traditional displays and driving a dimensional leap from planar/bendable to three-dimensional (3D)/stretchable. Thereby, they have unveiled vast application potential in wearable imaging systems, implanted phototherapy devices, and biomimetic electronic skin^{1–3}. Catering to this burgeoning technological paradigm, various stretchable displays have been extensively explored and advanced, including quantum dot light-emitting diodes (QLEDs)⁴, perovskite light-emitting diodes (PeLEDs)⁵, light-emitting electrochemical cells (LECs)⁶, alternating current electroluminescent devices (ACELs)⁷, and organic light-emitting diodes (OLEDs)⁸. Among them, OLEDs have become the mainstream choice for commercial displays, owing to inherent advantages in high efficiency, low operating voltage, large-area scalability, and mechanical

flexibility^{9,10}. Currently, the active layer in commercial OLEDs is primarily constructed from small-molecule organic semiconductor (SMOS) materials, which exhibit superior color purity and high uniformity, while enabling precise multilayer stacking via vacuum evaporation^{11–13}. The luminous efficiency of such devices is typically one order of magnitude higher than that of polymer semiconductor-based counterparts, representing a pivotal driver for the large-scale commercialization of flexible OLEDs. Despite the extensive reports on flexible OLEDs, high-performance fully stretchable OLEDs remain scarce. Notably, the inherent material properties of SMOS that underpin the commercial success of flexible OLEDs pose a fundamental bottleneck for stretchable device development. Owing to the strong intermolecular π - π stacking interactions and the limited free volume within SMOS matrices, the resultant films are difficult to undergo plastic deformation under external mechanical stress, manifesting inherent brittleness with an elongation at break typically below

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1%^{14–16}. Consequently, the optoelectronic performance of SMOS-based OLEDs deteriorates drastically or even ceases to function under tensile strain.

Over the past several years, tremendous research efforts have been devoted to advancing stretchable OLEDs (Supplementary Table S1), and two dominant technical strategies have been developed to solve the brittleness issue of SMOS: (i) Geometrically stretchable OLEDs (*gs*-OLEDs), fabricated by transferring flexible OLEDs onto pre-stretched substrates followed by strain release to form an accordion-like wavy surface, or by using rigid islands connected with stress-relieving interconnectors^{17–25}; (ii) Intrinsically stretchable OLEDs (*is*-OLEDs), fabricated by mixing SMOS with elastomers or polymeric matrices to construct composite emissive layers, whose mechanical flexibility derives from elastic phases rather than pure small-molecule materials^{26–29}. The former allows for the use of conventional high-performance yet fragile organic materials, electrodes, and multilayer encapsulation technologies. However, such geometrically structured devices often involve trade-offs in visual uniformity and skin-like deformability³⁰. In contrast, the latter can realize full stretchability through the intrinsic ductility of materials, without the need for complex geometric designs, while simultaneously boasting visual uniformity, skin-conformability, and a simple fabrication process³¹. Nevertheless, the stretchability enhanced by incorporated elastomers and polymeric matrices is intrinsically constrained by the disruption of intermolecular π - π stacking in SMOS, which is an essential structural prerequisite for superior optoelectronic efficiency^{15,32}. This structural disruption leads to an inevitable efficiency-stretchability trade-off that conventional material blending strategies are incapable of alleviating. To date, the state-of-the-art SMOS-based *is*-OLEDs exhibit a maximum current efficiency (CE) of 47.9 cd/A and an external quantum efficiency (EQE) of 17.0%²⁶, which are markedly inferior to those of cutting-edge flexible OLEDs, posing a formidable barrier to large-scale commercialization. Moreover, such devices demonstrate a limited maximum mechanical stretchability of merely 60% and can withstand only 100 stretch-release cycles at 20% strain. Thus, there is an urgent need for innovative strategies that can synergistically balance the mechanical stretchability and optoelectronic performance of SMOS-based fully stretchable OLEDs.

Herein, based on an interfacial stress regulation strategy, we propose a type of nanocracked stretchable OLEDs (*ns*-OLEDs) featuring both high efficiency and excellent stretchability. This strategy is compatible with diverse brittle and non-elastic materials, enabling the fabrication of fully stretchable OLEDs. The core concept of this strategy trades the density of nanocracks for the device's spatial deformability, while utilizing multiple conductive paths retained between nanocracks to sustain efficient charge injection, transport, and recombination. This process is analogous to decomposing the entire OLED device into numerous microscale OLED subunits with nanoscale spacing during stretching, thus ensuring high performance of the device even under large deformations. The specific design is as follows: i) Interfacial buffer layers (IBL) with high adhesion and toughness are introduced at the upper and lower interfaces of the OLED functional layer to maximize the dissipation of external mechanical loads, thereby homogenizing and minimizing the stress exerted on the OLED functional layer. This facilitates precise regulation of crack initiation and propagation, ultimately forming a uniformly distributed nanocrack network in the OLED functional layer. ii) By leveraging Young's modulus and elongation at break difference between the asymmetric anode and cathode of the OLED, the interfacial stress distribution applied to the active layer is further modulated, inducing vertically penetrating nanocracks within the active layer. iii) An adaptive deformation layer (ADL) with high adhesion and elasticity is incorporated at the bottom of the IBL, which greatly enhances the device's strain tolerance. Based on this versatile strategy, the fabricated fully stretchable SMOS-based OLEDs achieve a maximum luminance of 16,050 cd/m², a

CE of up to 85.7 cd/A, and a low turn-on voltage of 2.5 V. Notably, the device maintains 83.2% of its initial current efficiency even under a maximum tensile strain of 250%, outperforming all previously reported stretchable light-emitting devices. More strikingly, this strategy enables the fabrication of OLEDs with multiple emission colors, manifesting substantial potential for high-performance full-color displays. This work establishes a universal platform for constructing high-performance fully stretchable OLEDs, thereby paving the way for the advancement of stretchable displays toward the commercialization of wearable and implantable electronic devices.

Results

Design concept

Small-molecule stretchable OLEDs typically suffer from random crack formation and efficiency degradation under mechanical deformation, which limits their practical application. Here, we propose an interfacial stress regulation strategy (Fig. 1a) to address this challenge. Unlike conventional strategies that aim to suppress cracks via blending elastomers or polymeric components (which often sacrifice device efficiency), our strategy employs active crack engineering, guiding controlled nanocrack formation instead of crack inhibition to realize the simultaneous achievement of high-performance and full-stretchable *ns*-OLEDs. To implement this strategy, the IBL was integrated at both the top and bottom of the OLED functional layer to achieve efficient stress dispersion. Specifically, this design can homogenize and minimize the mechanical stress imposed on the OLED functional layer, thus triggering the generation of nanocracks. Due to the direct contact between the cathode/anode and the IBL, non-penetrating nanocracks are formed in the cathode and anode (Supplementary Fig. S1). The multiple conductive paths retained between these non-penetrating nanocracks guarantee efficient charge injection during the stretching process. In contrast, the active layer does not have direct contact with the IBL. Accordingly, its cracking behavior is predominantly governed by the adjacent cathode and anode. We utilized the inherent brittleness of SMOS materials, characterized by high stress sensitivity and ready fracture under stretching. By designing asymmetric cathode and anode with distinct Young's moduli and elongation at break (Supplementary Fig. S2), we successfully achieved precise regulation of the interfacial stress within the active layer, thereby enabling targeted modulation of the crack density and morphology therein. This elaborate structural design facilitates the formation of uniform vertical penetrating nanocracks within the active layer while avoiding lateral dislocations, thereby ensuring efficient charge transport and recombination in the interconnected intact regions. This is similar to decomposing the entire OLED device into numerous microscale OLED subunits with nanoscale spacing. Consequently, the entire device retains high efficiency even under large stretching. To further enhance the mechanical robustness of the integrated system, an ADL was introduced at the bottom of the IBL to mitigate device stress via stress isolation and delay crack propagation, thereby enhancing the overall strain tolerance of the device. In brief, the IBL-mediated stress homogenization, asymmetric electrode-guided crack modulation, and ADL-enhanced stress isolation act synergistically to establish a hierarchical integrated stress management system, enabling the sequential mechanism of stress dispersion–crack regulation–stress buffering within a single integrated architecture.

In this design, a thiol-ester-based UV-curable adhesive^{33,34}, i.e., Norland Optical Adhesive 68 (NOA68), was strategically selected as the IBL, because it can respectively form Al–S bonds and C = S bonds with the cathode and anode, thus enabling the construction of a mechanically robust interlocked interface (Supplementary Fig. S3). In addition, this adhesive possesses excellent tensile properties (84%) and toughness (17.0 MJ/m³) (Supplementary Fig. S4), thus enabling it to fully absorb stress energy and achieve energy dissipation. Highly

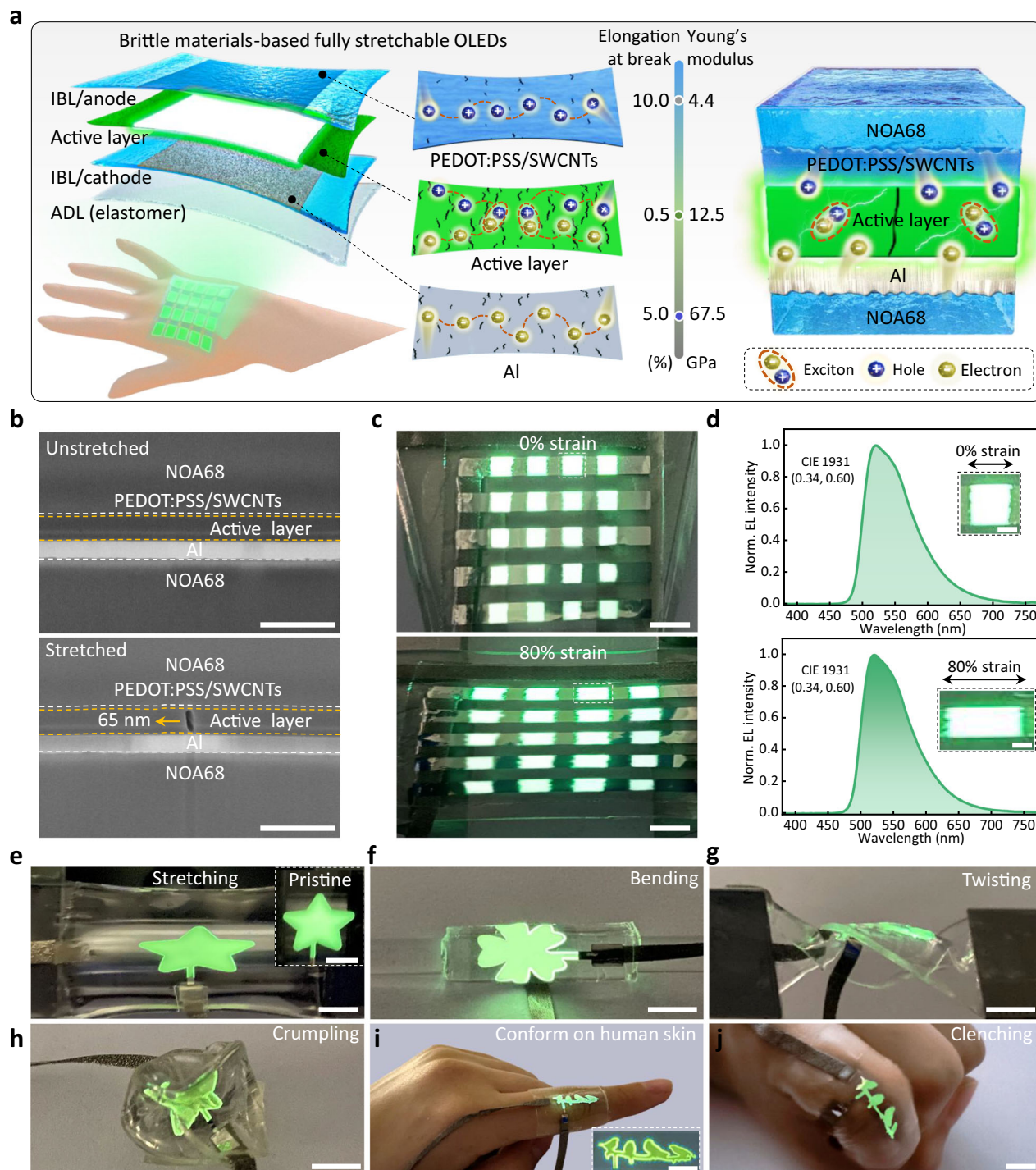


Fig. 1 | An interfacial stress regulation strategy for brittle materials-based ns-OLEDs. **a** Schematic of nanocracked stretchable OLEDs (ns-OLEDs), including adaptive deformation layer (ADL), interfacial buffer layers (IBL), polymer anode (PEDOT:PSS/SWCNTs), active layer (small-molecule organic semiconductors), and metal cathode (Al). **b–d** Focused ion beam scanning electron microscopy (FIB-SEM) images (**b**), photographs (**c**), and electroluminescence (EL) spectra with CIE

coordinates (**d**) of the 4×5 device array under 0% strain (top) and 80% strain (bottom). Scale bars, 500 nm in (**b**), 5 mm in (**c**), and 1 mm in the inset of (**d**). Representative images are shown; similar results were obtained from three independent experiments. **e–j** Photographs of the device under various deformation conditions, including stretching (**e**), bending (**f**), twisting (**g**), crumpling (**h**), conforming (**i**), and clenching (**j**). Scale bar, 5 mm.

transparent poly(3,4-ethylenedioxythiophene):poly(styrene sulfonate) (PEDOT:PSS)/single-wall carbon nanotubes (SWCNTs) and highly conductive aluminum (Al) serve as the anode and the cathode, respectively. On one hand, they possess suitable work functions to facilitate efficient charge injection. On the other hand, they exhibit significant differences in both Young's modulus and elongation at break, which is conducive to regulating the crack propagation

behavior in the active layer. Specifically, as shown in the middle panel of Fig. 1a, the Young's modulus and elongation at break of the cathode and anode are as follows: the anode is ≈ 4.4 GPa and $\approx 10\%$, while the cathode is ≈ 67.5 GPa and $\approx 5\%$. The active layer, with a relatively low Young's modulus (≈ 12.5 GPa), can form a Young's modulus gradient with the cathode and anode (Supplementary Fig. S5). Meanwhile, its low elongation at break ($\approx 0.5\%$) induced by intrinsic brittleness can be

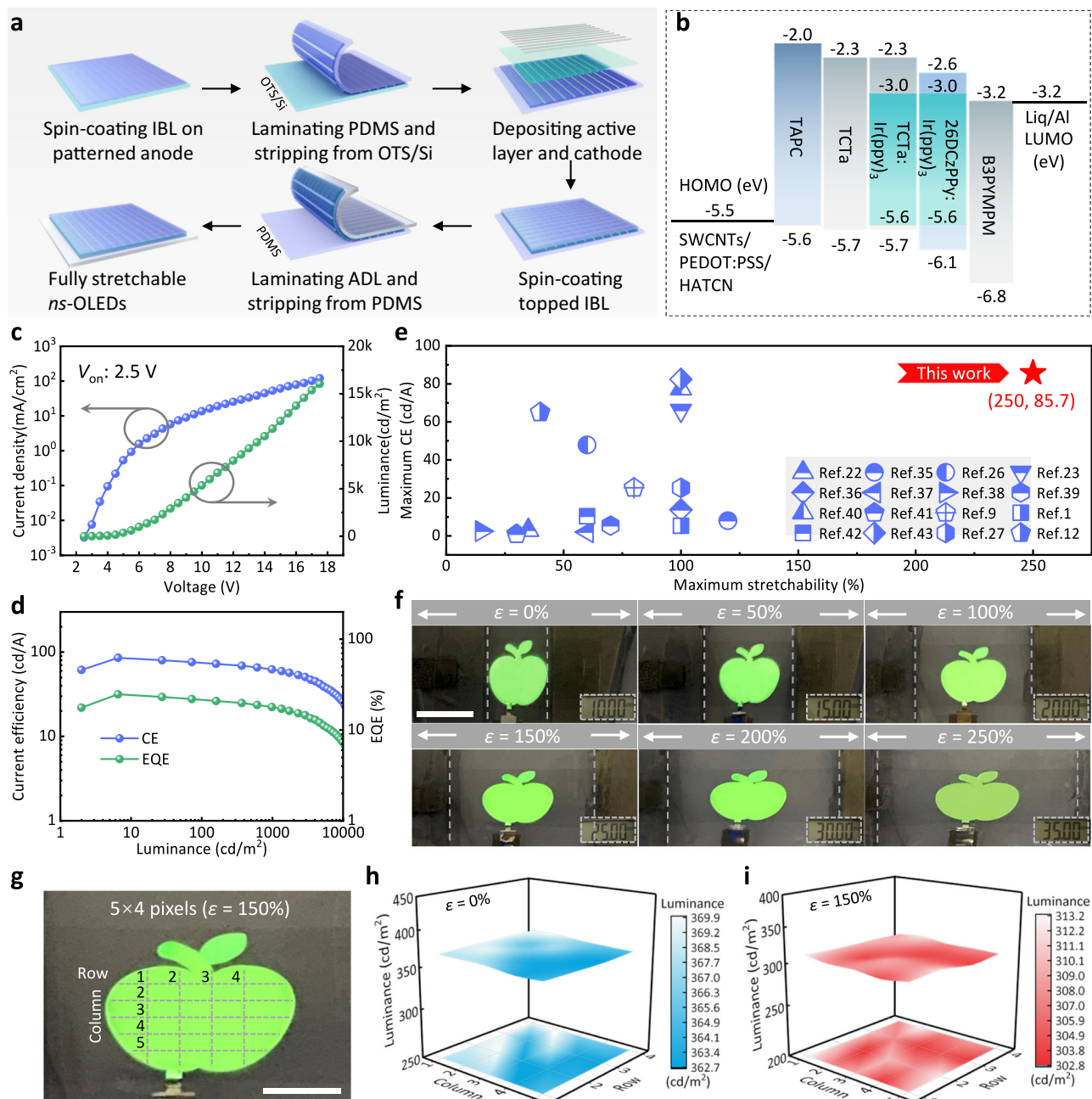


Fig. 2 | Fabrication strategy and optoelectronic performance. **a** Major fabrication scheme of fully stretchable ns-OLEDs. **b** Energy-level diagram of the device. **c**, **d** Electroluminescence (EL) performance under pristine condition. Current density–voltage–luminance (J–V–L) curves (**c**), current efficiency (CE), and external quantum efficiency (EQE) versus luminance curves (**d**). **e** Comparison of maximum

CE and stretchability in stretchable light-emitting diodes with previous reports. **f** Photographs of the device under tensile strains ranging from 0 to 250%. Scale bar, 10 mm. **g–i** Brightness uniformity of the device under 150% tensile strain. Photograph after stretching (**g**). Scale bar, 5 mm. Spatial distribution of brightness values before (**h**) and after (**i**) stretching.

regulated by the interfacial stresses from the two electrodes. The 3M-VHB 4918 elastomer was used as the ADL due to its high elongation at break (up to 860%), thus ensuring effective stress isolation and delaying crack propagation in the device under large deformations (Supplementary Fig. S6).

To intuitively verify the formation of nanocracks in the device during tensile deformation, cross-sectional specimens of both stretched and unstretched devices were fabricated via focused ion beam scanning electron microscopy (FIB-SEM). This characterization revealed robust interfacial adhesion across all functional layers enabled by our hierarchical integrated stress management system, and the exclusive formation of vertically penetrating nanocracks ($\approx 65 \text{ nm}$

in width) strictly confined to the active layer under tensile stress (Fig. 1b). Notably, the crack width is substantially smaller than the wavelength of visible emission, thereby precluding the formation of detectable optical scattering-induced dark stripes via classical diffraction effects. More importantly, the device preserves an intact vertical charge injection–transport–recombination pathway, as the nanocracks primarily divide the active layer into discrete microscale domains rather than interrupting the global vertical charge pathway. As shown in Fig. 1c, d, the 4×5 device array can be stretched to 80% strain while retaining an unshifted electroluminescence (EL) spectrum and stable Commission Internationale de l’Eclairage (CIE) coordinates of (0.34, 0.60). For a single large-area pixel ($10 \text{ mm} \times 10 \text{ mm}$), the

device can further sustain an overall tensile strain of up to 250% (corresponding to $\approx 50\%$ tensile strain in the emission region) without sacrificing functionality, which represents a significant improvement over state-of-the-art stretchable OLEDs (typically $<60\%$ maximum strain) and demonstrates the capability of ns-OLEDs to withstand tensile strains exceeding 200% while maintaining high performance (Fig. 1e). Strikingly, the device maintains reliable operation under diverse complex deformation scenarios: whether bent around a glass rod with a radius as small as 0.5 mm, crumpled into a spherical configuration, or conformally laminated onto the dynamically bending surface of a human finger (Supplementary Movie 1), its electrical performance remains largely uncompromised (Fig. 1f–j). These results highlight substantial potential for addressing the long-standing trade-off between high stretchability and operational stability in stretchable OLEDs, opening new avenues for next-generation wearable bioelectronics (e.g., real-time health monitoring devices integrated with conformal display modules), conformal human-machine interfaces, and dynamic curved displays, while breaking a critical bottleneck for the commercialization of stretchable optoelectronics.

Fabrication and performance characterization

Leveraging the aforementioned interfacial stress regulation strategy, we have successfully fabricated fully stretchable ns-OLEDs that synergistically achieve high efficiency and excellent stretchability. The main fabrication process of the device is illustrated in Fig. 2a. Briefly, an IBL was spin-coated onto the patterned PEDOT:PSS/SWCNTs anode, followed by ultraviolet (UV)-initiated curing. Subsequently, an elastic polydimethylsiloxane (PDMS) support layer was conformally laminated onto the IBL surface, enabling the detachment of the embedded transparent anode from the octadecyltrichlorosilane-modified silicon (OTS/Si) substrate via a gentle peeling process. It is worth noting that, compared with the PEDOT:PSS/SWCNTs anode prepared by the conventional layer-by-layer deposition process, the peeling-assisted approach significantly reduces the surface roughness of the anode. Further details are provided in Supplementary Note S1, Supplementary Figs. S7–12 and Supplementary Tables S2 and 3. The exfoliated transparent anode was then transferred into a high-vacuum thermal evaporation system, where the SMOS active layer and a metal cathode were deposited sequentially. Thereafter, a second IBL was spin-coated atop the device, followed by conformal lamination of an ADL. Finally, after peeling off the PDMS, a fully stretchable ns-OLED with a top-emission structure was obtained. Notably, the rational material selection and structural design of the device not only endow all functional layers with robust stretchability but also engineer an optimal energy-level alignment to facilitate efficient charge injection and transport (Fig. 2b). The specific device structure is as follows: SWCNTs (25 nm)/PEDOT:PSS (15 nm)/HATCN (10 nm)/TAPC (50 nm)/TCTa (10 nm)/TCTa: Ir(ppy)₃ (8 wt%, 10 nm)/26DCzPPy: Ir(ppy)₃ (8 wt%, 10 nm)/B3PYMPM (50 nm)/LiQ (1.5 nm)/Al (130 nm). Organic active layers consist of HATCN, TAPC, TCTa, and B3PYMPM as the hole injection layer (HIL), the hole transport layer (HTL), the electron blocking layer (EBL), and the electron transport layer (ETL). The *p*-type material TCTa and the bipolar material 26DCzPPy are respectively used as the host materials for the phosphorescent emitter Ir(ppy)₃. A defining hallmark of our fully stretchable ns-OLED is that its organic active layers are constructed entirely from commercially accessible materials. This intentional design departure from prevailing strategies that depend on custom-synthesized, scalability-limited materials. A central merit of this paradigm lies in its seamless compatibility with state-of-the-art high-performance materials and well-established fabrication technologies. This compatibility not only de-risks instrument development but also establishes a transformative pathway to close the critical chasm between lab-scale proof-of-concept and real-world commercialization.

The as-fabricated fully stretchable ns-OLED exhibits a low turn-on voltage of 2.5 V and a maximum brightness of 16,050 cd/m² (Fig. 2c). These performance metrics enable powering by commercial batteries, thereby meeting the stringent requirements for practical indoor lighting and display applications. Benefiting from the high-performance SMOS materials, the device exhibits high electroluminescent (EL) performance. The maximum CE and EQE reach 85.7 cd/A and 24.7%, respectively (Fig. 2d), outperforming all previously reported stretchable light-emitting diodes and rigid OLEDs based on the same material system (Fig. 2e, Supplementary Fig. S13, and Supplementary Table S1)^{1,9,12,22,23,26,27,35–43}. Not only that, the device maintains stable operation across diverse stretching states, even under a tensile strain of up to 250% (corresponding to a 250% expansion of the total electrode area and $\approx 50\%$ elongation of the emissive region), while retaining uniform brightness without significant degradation (Fig. 2f, Supplementary Fig. S14, and Supplementary Movie 2). Brightness uniformity stands as a critical prerequisite for the operational reliability of stretchable displays in practical deployment. Typically, SMOS-based stretchable devices undergo pronounced stress concentration upon direct stretching, triggering the formation of extensive microcracks or through-cracks. This not only induces insufficient luminance and undesirable visual artifacts but also compromises display fidelity and degrades user experience. Furthermore, it may even pose significant risks of electrical short circuits, ultimately leading to device failure. However, by leveraging our interfacial stress modulation strategy, we have successfully circumvented these long-standing bottlenecks. To quantify this performance, we systematically characterized the brightness uniformity of the device both in the pristine state and after stretching (Fig. 2g), evaluating pixel-level performance consistency through spatial brightness distribution mapping. As illustrated in Fig. 2h, i, the device retains high brightness uniformity across the entire emissive area in both states. Even under 150% tensile strain, the brightness maintains 84% of its initial value. Strikingly, this reveals that the formation of nanocracks in the functional layers exerts no significant impact on the device's optoelectronic performance or brightness uniformity. Collectively, these results demonstrate that our strategy effectively mitigates stress concentration at functional interfaces, enabling the device to retain high brightness uniformity even under large tensile strains. This breakthrough achievement overcomes the long-standing trade-off between high stretchability and superior EL efficiency in SMOS-based OLEDs, thereby paving a viable pathway toward the commercialization of next-generation stretchable display technologies.

Impact of architecture optimization on device performance

The device architecture serves a critical role in modulating the initiation and propagation of nanocracks. To elucidate how structural disparities influence the mechanical robustness of stretchable SMOS-based OLEDs, we fabricated three device configurations. Figure 3a presents schematic illustrations and 3D digital microscope images of the three devices under varying tensile strains. Herein, FL denotes the number of device' functional layers, encompassing the anode, active layer, and cathode. For the ADL + FL structure, prominent cracks with widths of tens of micrometers emerge on the surface when the device is stretched to 40% strain, which further propagate and widen with increasing tensile strain. At 40% strain, the area fraction of microcracks reaches 46.4%, and increases further to 55.8% at 80% strain (Supplementary Fig. S15). This phenomenon stems primarily from the substantial mismatch in Young's modulus between the ADL (15.6 kPa) and FL (33.6 GPa). While the elastic ADL exhibits high stretchability, the brittle FL cannot accommodate the deformation induced during stretching. FIB-SEM images in Fig. 3b further reveal that all functional layers are fully fractured with cracks propagating through the entire device thickness. Furthermore, finite element analysis (FEA) simulations confirm that the maximum stress in the central region of the

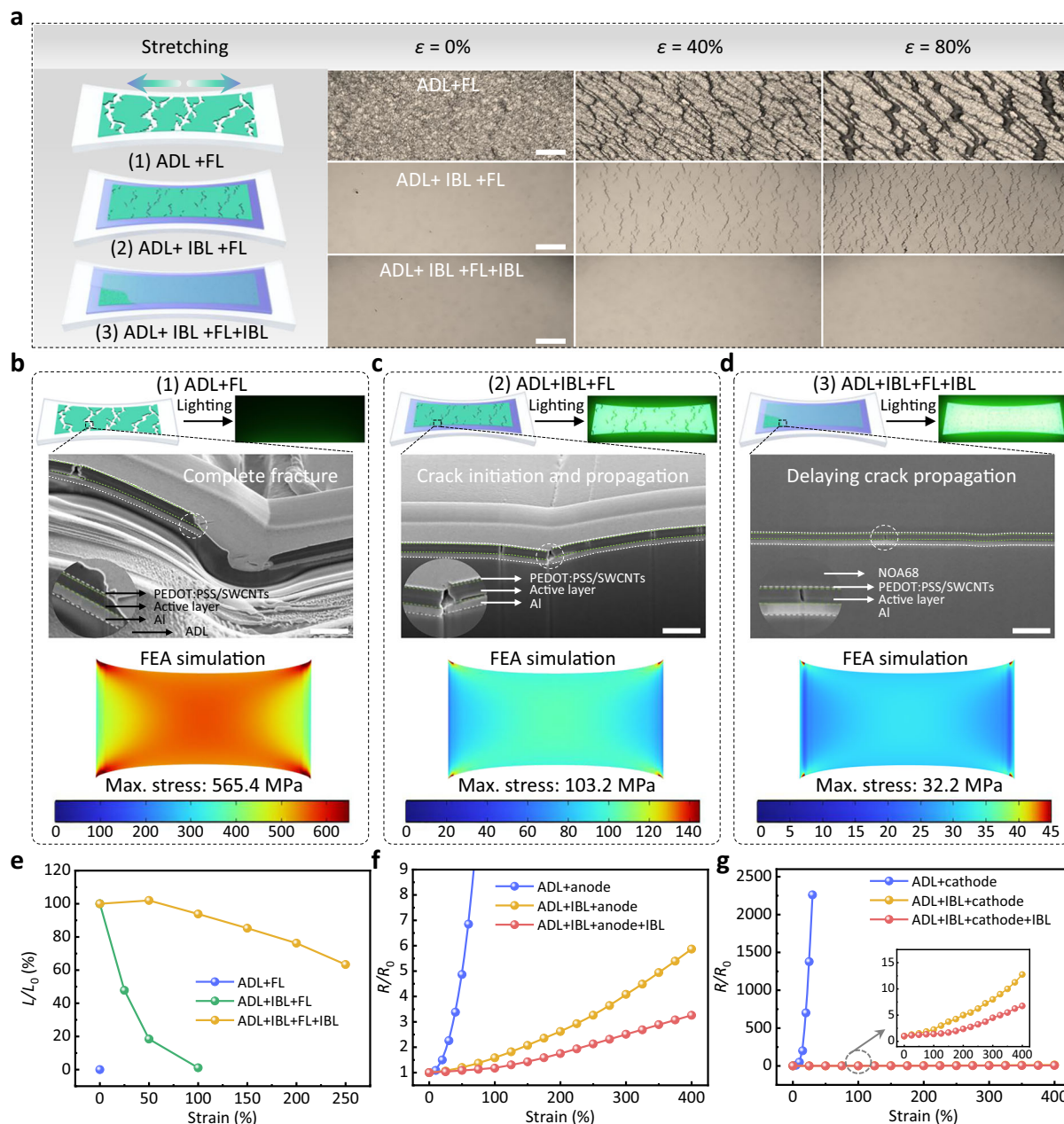


Fig. 3 | Influence of device architecture on mechanical performance.

a Schematic illustrations and three-dimensional (3D) optical microscope images of three device architectures (ADL + FL, ADL + IBL + FL, ADL + IBL + FL + IBL) under tensile strains ranging from 0 to 80%. ADL represents the adaptive deformation layer, IBL represents the interfacial buffer layer, and FL represents the functional layer of the device, including the anode, active layer, and cathode. Scale bar, 100 μm . **b–d** Cross-sectional FIB-SEM images of the three devices at 80% tensile strain, with insets showing magnified views of localized crack morphologies. Scale bar, 1 μm . Representative images are shown; similar results were obtained from

three independent experiments. Top panels: schematic diagrams of the respective device architectures after tensile deformation. Bottom panels: corresponding finite element analysis (FEA) simulations of the stress distribution within each device under 80% tensile strain. **b** Baseline ADL + FL structure. **c** Single-IBL ADL + IBL + FL structure. **d** Dual-IBL ADL + IBL + FL + IBL structure. **e** Normalized luminance variation curves of the three devices as a function of tensile strain from 0 to 250%. **f, g** Normalized resistance (R/R_0) variation curves of the device electrodes under tensile strains ranging from 0 to 400%, for the anode (**f**) and cathode (**g**).

device reaches 565.4 MPa, indicating that severe stress concentration is likely the key driver for crack propagation. After an IBL was inserted between the ADL and FL, the crack width was significantly reduced. This is attributed to the IBL's moderate Young's modulus (127.4 MPa), which lies between those of the ADL and FL, thereby effectively mitigating crack propagation (Fig. 3c). FEA simulations further verify that the maximum stress in the central region of this modified structure decreases to 103.2 MPa. However, as multiple organic functional layers

and the Al electrode are stacked atop the ADL and IBL, these upper layers lack adequate mechanical protection during stretching, leading to the formation of numerous microcracks on the device surface.

By contrast, the architecture incorporating an IBL atop the FL markedly alleviates these mechanical failure issues. The dual-IBL configuration effectively encapsulates and safeguards the device's functional layers, forming robust interfacial adhesion and thereby preserving excellent mechanical stability under high tensile strains.

FEA simulations further demonstrated that the maximum stress in the central region of this optimized structure plummeted to 32.2 MPa (Fig. 3d), which is consistent with the experimentally observed delayed crack propagation. Critically, no microcracks were detected on the device surface even at 80% tensile strain; only localized nanocracks (≈ 65 nm in width) emerged within the active layer. Furthermore, under more complex biaxial stretching states, this optimized architecture consistently suppresses large-scale crack propagation and confines nanocracks within the active layer at strains up to 80%. A detailed investigation of the stress distribution, crack evolution, and EL performance under biaxial deformation is provided in Supplementary Note S2 and Supplementary Figs. S16–19. In our experiments, these sub-wavelength-scale nanocracks do not compromise macroscopic EL performance for two core reasons: first, the visible emission wavelength of the OLED (400–700 nm) is one order of magnitude larger than the crack width, precluding the formation of detectable optical scattering-induced dark stripes via classical diffraction effects; second, the device's vertical charge injection-transport-recombination paradigm remains intact, as the cracks merely partition the active layer rather than disrupting the global vertical charge pathway. In this context, this is analogous to partitioning the OLED into micron-scale subunits separated by nanocracks, each retaining efficient carrier injection, transport, and exciton recombination. As illustrated in Fig. 3e, the ADL + FL architecture suffered from severe stress-concentration-induced cracking, leading to catastrophic functional layer failure and thus complete loss of optoelectronic performance. For the ADL + IBL + FL structure, the luminance declined drastically with increasing strain, with electrical functionality almost entirely lost at 100% strain. Conversely, the ADL + IBL + FL + IBL architecture exhibited extraordinary mechanical stretchability, retaining 63% of its initial luminance even under a high tensile strain of 250%. Further, the anode and cathode, respectively, maintained low normalized resistance values (R/R_0) of only 3.3 and 6.8, even at an ultrahigh tensile strain of 400% (Fig. 3f, g). Such electrode stability, which is unmatched by single-IBL or baseline configurations, acts as a pivotal safeguard for sustaining unimpeded charge injection under large mechanical deformation, directly underpinning the marked improvements in overall device performance.

Impact of IBL properties on device performance

As the IBL in fully stretchable ns-OLEDs is constructed from UV-curable polymers, its bulk and interfacial mechanical properties are exquisitely sensitive to UV irradiation duration. This is a critical processing parameter that dictates the overall mechanical resilience of the integrated device (Fig. 4a). Figure 4b displays 3D optical microscope images of dual-IBL ns-OLEDs under tensile deformation as a function of IBL UV irradiating time. At a short irradiating time of 20 s, pronounced microcracks emerge across the device surface at tensile strains over 80%, compromising structural integrity. By contrast, when the irradiating time is optimized to 140 s, the device retains an almost defect-free surface under 80% strain, with only isolated fine cracks observable even at 200% strain, indicative of superior mechanical tolerance. However, further extending the irradiating time to 220 s triggers a marked escalation in crack density and propagation rate. At 80% tensile strain, cracks begin to emerge across the entire device area, which negates the mechanical advantages afforded by the IBL architecture.

To elucidate the origin of this irradiation-dependent mechanical behavior, we probed the intrinsic mechanical properties of UV-curable IBL films via uniaxial tensile testing (Supplementary Fig. S20), with key metrics (elongation at break, toughness, Young's modulus) summarized in Fig. 4c, d. As irradiating time increases from 20 to 140 s, the IBL's elongation at break rises from 32 to 84%, toughness surges from 0.7 to 17.0 MJ/m³, and Young's modulus increases significantly from 15.6 to 127.4 MPa. This concurrent improvement in ductility, toughness, and stiffness stems from the formation of a moderately

crosslinked three-dimensional polymer network. Controlled UV exposure facilitates the assembly of a dense yet non-overconstrained network, where covalent crosslinks enhance structural integrity while retaining adequate molecular chain mobility to dissipate stress through chain slippage and segmental rotation^{44,45}. By contrast, irradiating time exceeding 140 s induces over-cross-linking, which immobilizes polymer chains into a rigid, brittle network with severely constrained segmental mobility. As a result, elongation at break plummets to 55%, toughness drops to 15.7 MJ/m³, and Young's modulus increases sharply to 311.5 MPa, which is a mechanical profile inconsistent with the compliance demands of highly stretchable OLEDs. This over-cross-linking effect abrogates the network's capacity to relieve applied stress via chain dynamics, concentrating stress at network junctions and generating intrinsic fracture initiation sites that propagate rapidly under tensile loading⁴⁶. Critically, these bulk mechanical trends correlate directly with device-level performance. The 140 s-irradiated IBL attains an optimal balance of stiffness and ductility, enabling it to mitigate the modulus mismatch between the ADL and FL (as established in Fig. 3) and suppress crack initiation and propagation, whereas under- or over-irradiated IBLs either lack structural rigidity or exhibit insufficient ductility to preclude mechanical failure. As illustrated in Fig. 4e and Supplementary Fig. S21, the device irradiated for 140 s retains 63.4% of its initial brightness and 83.2% of its initial current efficiency, even at a tensile strain of 250%. Furthermore, FEA simulations confirm that the 140 s-irradiated device exhibits the lowest maximum stress (32.2 MPa) in the central region (Fig. 4f).

Apart from that, the thickness of the IBL has a significant effect on the mechanical performance of the device. As the IBL thickness increases from 0 to 20 μ m, the area fraction of microcracks decreases markedly (Fig. 4g). This improvement is attributed to the stronger deformation absorption capabilities of the relatively thicker IBL, which can alleviate external tensile stress through its own elastic deformation, thereby reducing excessive stress concentration at local regions or interfaces. As displayed in Fig. 4h, the device without an IBL loses its EL performance due to the stress concentration from the underlying ADL, whereas the device with a 20 μ m-thick IBL exhibits more stable luminance changes, maintaining 76.2% of its initial brightness even under 200% strain. Furthermore, after 1500 stretch–release cycles, the device retains 86.5% of its initial brightness under uniaxial stretching at 30% strain (Fig. 4i) and 81.2% under biaxial stretching at 20% strain (Supplementary Fig. S22). The detailed crack morphology and distribution after repeated stretch–release cycles are further discussed in Supplementary Note S3 and Supplementary Figs. S23–26. This cyclic resilience stems from the IBL's fatigue-resistant cross-linking network, which prevents interfacial delamination during repeated stretch–recovery cycles. Given the equally critical role of bending reliability in stretchable and skin-like display applications, we further assessed the device performance under progressively reduced bending radii and repeated bending–recovery cycles. During consecutive bending from a radius of 6 to 1 mm, the device retained 90.4% of its initial brightness at the minimum radius of 1 mm. Furthermore, after 10,000 bending–recovery cycles at a bending radius of 5 mm, the device maintained 80.4% of its initial brightness, demonstrating excellent mechanical reliability under bending deformation (Supplementary Fig. S27).

Synergistic modulation of nanocracks by anode and cathode

Building on the asymmetric electrode design with tailored Young's moduli and elongations at break for the cathode and anode (as detailed earlier), we further demonstrate its efficacy in precise modulation of interfacial stress within the SMOS active layer, enabling targeted control over the density and morphology of nanocracks in the active layer. To validate how this asymmetric configuration governs nanocrack initiation and propagation, we systematically investigated three device architectures with distinct modulus and elongation at

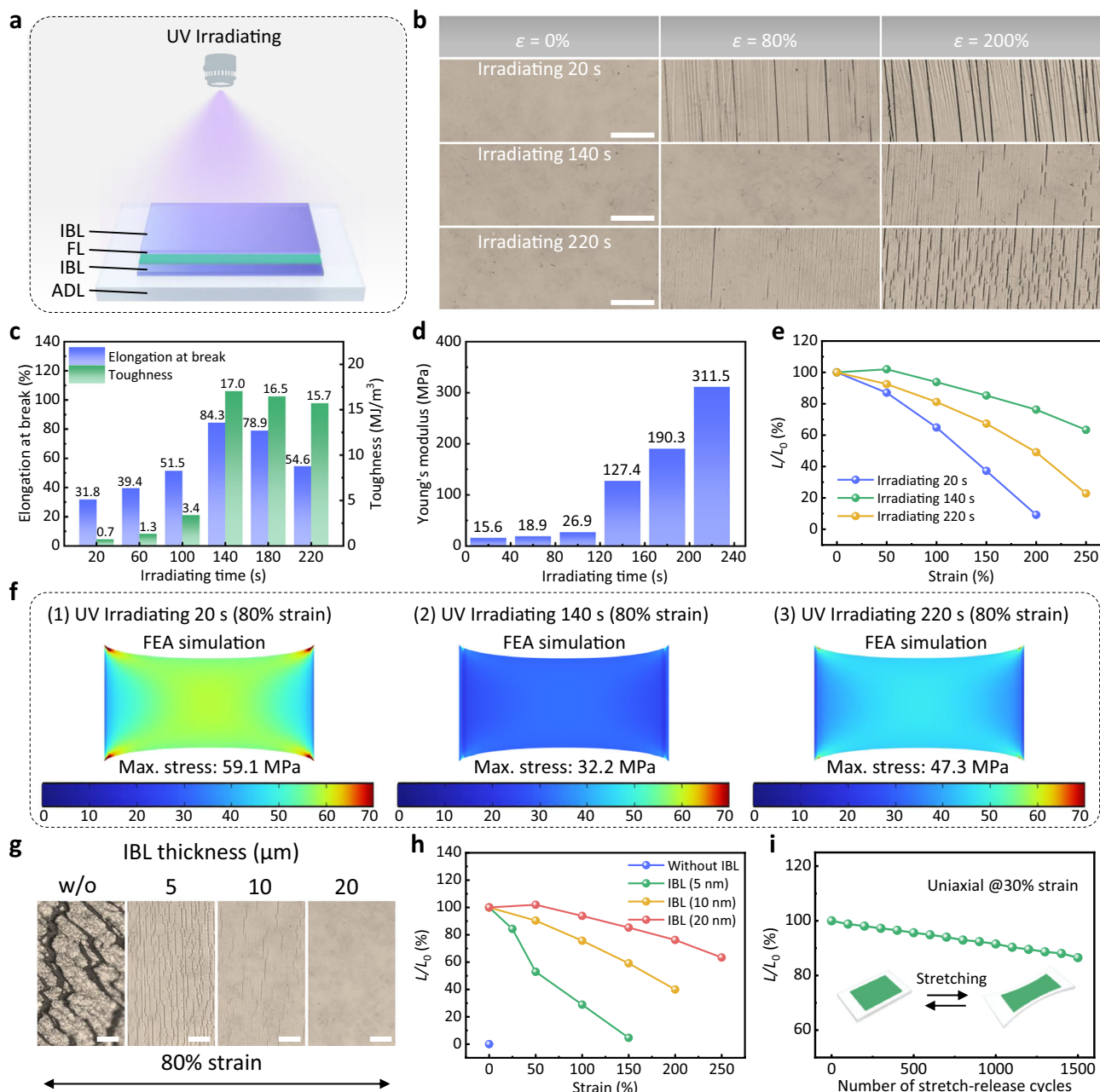


Fig. 4 | Influence of interfacial buffer layer (IBL) properties on device performance. **a** Schematic illustration of ultraviolet (UV) irradiation on the IBL. **b** 3D optical microscopy images of the devices at tensile strains of 0%, 80%, and 200% under different irradiating times. Scale bar, 50 μm . **c**, **d** Changes in elongation at break, toughness (**c**), and Young's modulus (**d**) of IBL under different irradiating times. **e** Normalized luminance variation curves of the devices under tensile strains from 0 to 250% at different irradiating times. **f** Finite element analysis (FEA)

simulation of the stress distribution in the device under 80% strain at different irradiating times. **g** 3D optical microscope images of the device under 80% strain for IBLs with thickness ranging from 0 to 20 μm . Scale bar, 50 μm . **h** Normalized luminance variation curves of the devices under tensile strains from 0 to 250% at different IBL thicknesses. **i** Normalized luminance variation of the device during repeated stretch-release cycles under uniaxial stretching at 30% strain.

break distributions (Fig. 5a–c). As depicted in Fig. 5a, when the SMOS active layer is sandwiched between two rigid Al electrodes (symmetric high-modulus configuration), numerous vertically penetrating nano-cracks are generated within the active layer under 80% tensile strain. This observation stems from the combined effect of substantial mechanical incompatibility in both Young's modulus and elongation at break: Al exhibits a Young's modulus of ≈ 67.5 GPa and an elongation at break of $\approx 5\%$, whereas the SMOS active layer has a Young's modulus of ≈ 12.5 GPa and an ultra-low elongation at break ($< 1\%$). This dual mismatch induces intense stress concentration at the electrode/active layer interfaces. Since the rigid Al electrodes cannot accommodate the

tensile deformation of the SMOS layer, nor can the brittle SMOS layer withstand the stress transferred from the Al. The large interfacial stress gradient serves as preferential nucleation sites for cracks, which subsequently propagate vertically through the brittle SMOS active layer due to the absence of effective stress dissipation pathways in the rigid symmetric system.

In contrast, when the SMOS active layer is sandwiched between two flexible PEDOT:PSS/SWCNTs electrodes (symmetric low-modulus configuration), no interfacial cracks are detected (Fig. 5b). Instead, a high density of diffuse, randomly oriented circular cracks forms within the active layer. This behavior originates from the matched Young's

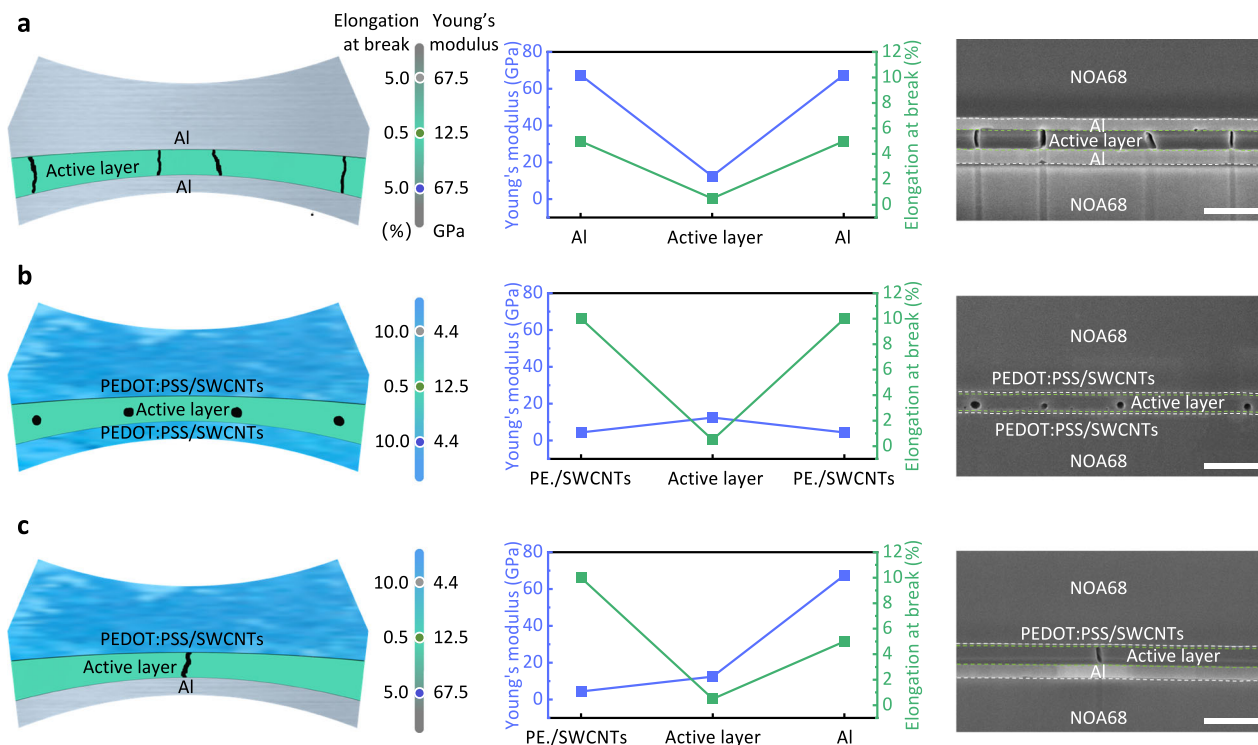


Fig. 5 | Nanocrack regulation by the anode and cathode. **a–c** Schematic illustrations, Young's modulus and elongation at break data (PE. denotes PEDOT:PSS), and FIB-SEM images of three device configurations under 80% tensile strain. Scale bar, 500 nm. Representative images are shown; similar results were obtained from

three independent experiments. **a** Al/active layer/Al symmetric high-modulus configuration. **b** PEDOT:PSS/SWCNTs/active layer/PEDOT:PSS/SWCNTs symmetric low-modulus configuration. **c** PEDOT:PSS/SWCNTs/active layer/Al asymmetric gradient modulus configuration.

modulus (PEDOT:PSS/SWCNTs ≈ 4.4 GPa vs. SMOS ≈ 12.5 GPa) and the superior elongation at break of PEDOT:PSS/SWCNTs ($\approx 10\%$), which enables the flexible electrodes to conformally follow the tensile deformation of the SMOS layer. This compliance efficiently mitigates interfacial stress concentration, suppressing interfacial crack initiation. However, owing to the ultra-low elongation at break of SMOS ($<1\%$), the applied tensile stress is inevitably transferred and concentrated within the active layer. Although the flexible electrodes effectively alleviate interfacial stress concentration, they cannot dissipate the tensile stress imposed on the brittle SMOS layer itself. Without viable stress dissipation pathways inside the active layer, the concentrated internal stress ultimately triggers the formation of random circular cracks.

Noticeably, the asymmetric gradient modulus architecture (low-modulus PEDOT:PSS/SWCNTs top electrode and high-modulus Al bottom electrode) yields a dramatic reduction in nanocrack density (Fig. 5c). The synergistic regulation of Young's modulus and elongation at break in this asymmetric design is critical for crack control: the PEDOT:PSS/SWCNTs top electrode (≈ 4.4 GPa, $\approx 10\%$ elongation at break) accommodates partial tensile deformation to reduce interfacial stress, while the Al bottom electrode (≈ 67.5 GPa, $\approx 5\%$ elongation at break) provides mechanical constraint to guide crack propagation. In this configuration, the mechanical gradient introduced by the asymmetric electrodes dictates nanocrack formation, with interfacial stress efficiently dissipated through the Al/active layer interface. Compared to the symmetric dual-Al electrode configuration, the reduction in rigid interfaces (from two to one) minimizes the number of stress concentration hotspots, leading to a substantial decrease in crack initiation. Supplementary Fig. S28 additionally demonstrates that the asymmetric gradient modulus architecture achieves a crack spacing of ≈ 4.2 μm , which is substantially larger than those of the symmetric rigid (≈ 0.5 – 0.8 μm) and symmetric low-modulus (≈ 0.5 – 1.0 μm) structures. Furthermore, leveraging the stress-sensitive nature of SMOS and the

directional stress guidance from the asymmetric elongation at break distribution, nanocracks preferentially propagate along the vertical direction rather than the lateral direction, avoiding the formation of horizontal dislocations that would sever charge transport pathways. The practical significance of this electrode design was further clarified by comparison with the conventional ITO/active layer/Al architecture, as detailed in Supplementary Note S4 and Supplementary Figs. S13 and 29. In addition, the acceptable design window of the electrode modulus-gradient architecture was further clarified by systematically varying the anode and cathode moduli, demonstrating that the crack-guiding behavior is not limited to a single modulus combination but originates from the asymmetric mechanical boundary between a low-modulus anode and a high-modulus metal cathode, as detailed in Supplementary Note S5 and Supplementary Figs. S7–9, 30–33. All in all, these results demonstrate that the tailored combination of Young's modulus gradient and asymmetric elongation at break of the cathode and anode enables precise nanocrack regulation. It facilitates strain dissipation via a limited number of vertical nanocracks, mitigates stress concentration in the brittle SMOS active layer, and preserves the structural integrity essential for stable optoelectronic performance under large mechanical deformations. These findings clarify the synergistic role of electrode modulus and elongation at break in governing crack behavior in stretchable devices.

Universal applicability of our strategy to multicolor ns-OLEDs

The proposed interfacial stress regulation strategy exhibits broad applicability to fully stretchable SMOS-based OLEDs across multiple emission colors. To confirm this universality, we investigated three representative small-molecule organic emitters: the blue-emitting material Firpic, the yellow-emitting material PO-01, and the red-emitting material Ir(piq)₃. The energy-level diagrams and molecular structures of these materials are provided in Supplementary Fig. S34. As illustrated in Fig. 6a, the main emission peaks of the

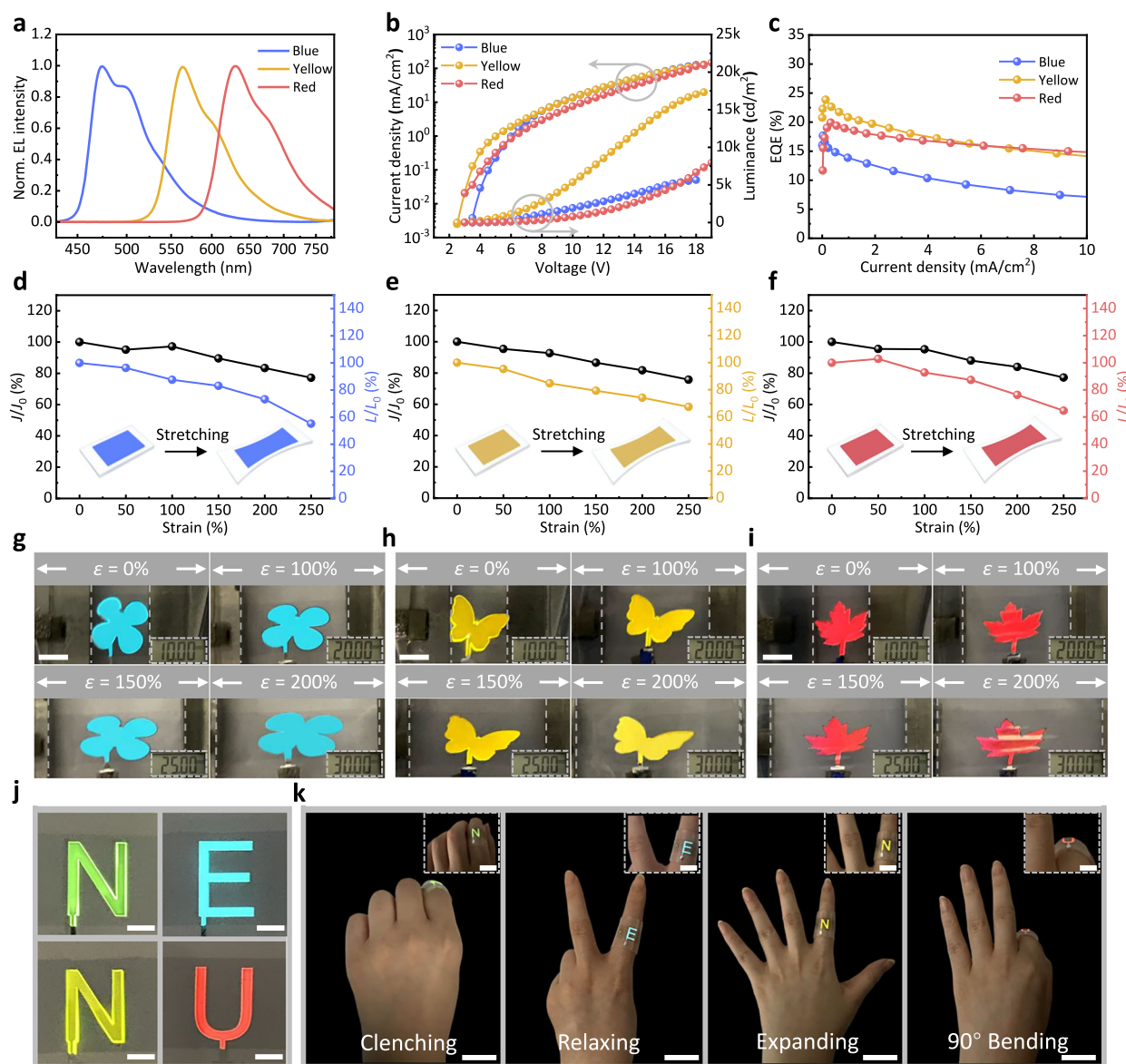


Fig. 6 | Universal strategy for multicolor stretchable small-molecule organic semiconductor (SMOS)-based OLEDs. **a–c** Electroluminescence (EL) performance of fully stretchable blue, yellow, and red SMOS-based OLEDs. **a** EL spectrum. **b** Current density–voltage–luminance (J–V–L) curves. **c** External quantum efficiency (EQE)–current density curves. **d–f** Variation curves of current density and luminance for the three devices under tensile strains ranging from 0 to 250%. **d** Blue-emitting device. **e** Yellow-emitting device. **f** Red-emitting device. **g–i** Digital

photographs of three devices with different colors and shapes under tensile strains ranging from 0% to 200%. Scale bar, 5 mm. **g** Blue-emitting device. **h** Yellow-emitting device. **i** Red-emitting device. **j** Digital photographs of emission patterns in four colors. Scale bar, 2 mm. **k** Digital photographs of fingers performing various movements and postures, such as clenching, relaxing, expanding, or bending at 90°. Scale bar, 2 cm. Insets show magnified views of the emitting devices. Scale bar, 1 cm.

corresponding devices are centered at 472 nm (Firpic), 560 nm (PO-01), and 628 nm (Ir(piq)₃), confirming bright and pure blue, yellow, and red electroluminescence. Further characterizations of the current density–voltage–luminance (J–V–L) and EQE–L characteristics (Fig. 6b, c) reveal excellent optoelectronic performance. The blue device exhibits a turn-on voltage of 3.5 V, maximum luminance of 5660 cd/m², and peak EQE of 17.7%; the yellow device shows 2.5 V, 17,320 cd/m², and 23.9%; and the red device achieves 3.0 V, 7940 cd/m², and 19.9%. Importantly, the overall performance of these stretchable devices is comparable to that of state-of-the-art rigid or flexible OLEDs based on the same emissive systems^{47–49}, highlighting the compatibility of our strategy with high-performance organic emissive materials.

For practical wearable and conformable optoelectronic applications, we systematically assessed the mechanical robustness and operational stability of the multicolor ns-OLEDs. As shown in Fig. 6d–f, after stretching to 100% strain, the current density and luminance of all three devices retained over 90% and 80% of their initial values, respectively. Moreover, patterned blue, yellow, and red ns-OLEDs maintained normal operation even under 200% strain, with uniform electroluminescence preserved (Fig. 6g–i). Crucially, the multicolor ns-OLEDs demonstrate excellent operational endurance under repeated mechanical deformation, retaining over 80% of their initial luminance after 1500 stretch-release cycles to 30% tensile strain (Supplementary Fig. S35). To simulate real-world dynamic scenarios, four-color ns-OLEDs were conformally laminated onto finger joints (Fig. 6j–k). During repeated finger movements,

including clenching, relaxing, spreading, and 90° bending, all color devices exhibited stable electroluminescence throughout the entire deformation process. Overall, these results demonstrate that our proposed strategy is compatible with a broad spectrum of small-molecule semiconductor materials and conventional vacuum deposition processes for rigid/flexible OLEDs, thereby achieving unprecedented universality for multicolor fully stretchable SMOS-based OLEDs. This work not only overcomes the long-standing mechanical fragility bottleneck of brittle organic semiconductors in stretchable optoelectronics but also lays a solid foundation for the development of next-generation wearable, conformable, and high-performance multicolor optoelectronic devices.

Discussion

In conclusion, our interfacial stress regulation strategy provides a generalizable design paradigm for overcoming the mechanical fragility of brittle semiconductors in stretchable devices. By leveraging active crack engineering and a hierarchical stress management system, we have achieved the synergistic integration of high efficiency and ultra-high stretchability in ns-OLEDs, with broad multicolor universality and compatibility with commercial manufacturing processes. On this basis, the fabricated fully stretchable ns-OLEDs achieved a maximum luminance of 16,050 cd/m², a current efficiency of up to 85.7 cd/A, and a low turn-on voltage of 2.5 V. Besides, the device can withstand a tensile strain of up to 250% while maintaining 83.2% of its initial current efficiency, outperforming all previously reported stretchable light-emitting devices. This work not only breaks the long-standing trade-off between stretchability and efficiency but also redefines the existing paradigm of stretchable OLEDs, accelerating the translation of stretchable optoelectronics from academic research to practical applications and paving the way for next-generation wearable bioelectronics, conformal human-machine interfaces, and dynamic curved displays.

Methods

Ethics statement

The experiment involving a human participant was conducted in accordance with all relevant ethical regulations. The study protocol was approved by the Science and Technology Ethics Committee of Northeast Normal University (Authorization number: 202602003). Written informed consent was obtained from the participant prior to the experiment.

Human participant information

The skin-attachment demonstration experiment was performed by the first author as a volunteer participant (male, 30 years old). Sex was self-reported and was not considered as a variable in the study design because the experiment was intended solely to demonstrate device conformability and functionality. No compensation was provided for participation.

Materials

Transparent bottom anodes, including PEDOT:PSS (Clevios PH1000), ethylene glycol (EG, 99%), and surface-active agent (Capstone FS-30, 98%) were purchased from Heraeus, DuPont, and Sigma-Aldrich, respectively. SWCNTs aqueous dispersion with the initial concentration of 0.22 mg/ml was procured from Chengdu Organic Chemicals Co., Ltd., Chinese Academy of Sciences. NOA68 was provided by Norland Corporation. 3M-VHB 4918 was purchased from Minnesota Mining and Manufacturing (3M) Company. PDMS (Sylgard 184) was acquired from Dow Corning Corporation. Octadecyltrichlorosilane (OTS, 95%) solution was purchased from Acros Co., Ltd. Acetone, isopropanol, chloroform, and concentrated nitric acid were purchased from Sigma-Aldrich. Photoresist (AZ 5200NJ) and developer (AZ 400 K) were purchased from Merck. Small-molecule organic

semiconductors including 1,4,5,8,9,11-hexaazatriphenylenehexacarboxitrile (HATCN), 2,6-bis(3-(carbazol-9-yl)phenyl)pyridine (26DCzPPy), 1-Bis[4-[N,N-di(4-tolyl)amino]phenyl]-cyclohexane (TAPC), Tris(4-carbazoyl-9-ylphenyl)amine (TCTa), 4,4'-bis(N-carbazoyl)biphenyl (CBP), 3,3'-di(9H-carbazol-9-yl)-1,1'-biphenyl (mCBP), Bis-4,6-(3,5-di-3-pyridylphenyl)-2-methylpyrimidine (B3PYMPM), Bis[2-(4,6-difluorophenyl)pyridyl-N,C2]iridium(III) Picolate (Firpic), Tris[2-phenylpyridine]iridium(III) (Ir(ppy)₃), 2,7-Bis(2,2'-bipyridin-6-yl)triphenylene (PO-01), Tris(1-phenylisoquinoline)iridium(III) (Ir(piq)₃), 8-Quinolinolato lithium (Liq) were purchased from Luminescence Technology Corp. and Xi'an Polymer Light Technology Corp.

Fabrication of transparent polymer anode

(I) The Si substrate was sequentially cleaned with acetone, isopropanol, and deionized water for 20 min, then dried with nitrogen gas and further dried in an oven at 130 °C for 15 min. Subsequently, the Si substrate was hydroxylated using oxygen plasma treatment (100 W, 3 min), followed by immersion in a hydrophobic OTS solution (OTS: heptane = 1:1000) for 7 min to modify the surface interface and reduce surface adhesion, thus enabling facile delamination of the device from the Si substrates in later steps. Next, the OTS/Si substrate was immersed in chloroform for 10 min and dried with N₂. (II) The PEDOT:PSS solution, prepared by adding 6% EG to the original PH1000 to enhance conductivity and 0.2% FS-30 to improve interfacial wettability, was spin-coated onto the OTS/Si substrate at 6000 rpm for 30 s. The resulting film was annealed at 80 °C for 60 min, followed by immersion in concentrated nitric acid for 3 min to remove insulating PSS chains, thereby further improving its electrical conductivity. After drying with N₂, the OTS/Si substrate coated with the PEDOT:PSS film was placed onto the hotplate (160 °C) while an aqueous SWCNTs dispersion (0.15 mg/mL) was sprayed onto it, followed by a second immersion in concentrated nitric acid for 3 min and subsequent drying with N₂. (III) The PEDOT:PSS/SWCNTs anode was patterned using a conventional photolithography process. The photoresist (AZ 5200NJ) was spin-coated onto the anode. The pattern was defined sequentially by UV exposure (μMLA 100, Heidelberg Instruments), development (in AZ 400 K), and fixing (in deionized water for 3 s). Afterward, the exposed PEDOT:PSS/SWCNTs regions were etched by oxygen plasma for 1.5 min under conditions of 0.1 MPa oxygen content, 7 sccm flow rate, 0.34 mbar pressure, and 100 W power. Next, the residual photoresist was removed by dissolving it in acetone. Finally, the Si substrate exposed to oxygen plasma was re-treated with an OTS solution to ensure uniform interfacial modification.

Fabrication of ns-OLEDs

(I) The IBL (NOA68) was spin-coated onto the patterned PEDOT:PSS/SWCNTs anode at 4000 rpm for 30 s and then precured under 365 nm UV irradiation with a 300 W lamp for 10 s. (II) An elastic support layer (PDMS, cross-linking ratio of 10:1) was mechanically laminated onto the surface of the NOA68, the embedded transparent anode with PDMS could be non-destructively stripped from OTS/Si substrate, freeing it from the mechanical constraints of the rigid substrate, thus obtaining a flexible transparent anode. (III) The stripped anode was transferred into a vacuum thermal evaporation chamber under a pressure of 1×10^{-4} Pa, where small-molecule organic semiconductors and the metal cathode were sequentially deposited. The layers were deposited in the following order: For the green-emitting device, the structure was HATCN (10 nm)/TAPC (50 nm)/TCTa (10 nm)/TCTa: Ir(ppy)₃ (8 wt%, 10 nm)/26DCzPPy: Ir(ppy)₃ (8 wt%, 10 nm)/B3PYMPM (50 nm)/Liq (1.5 nm)/Al (130 nm). For the blue-emitting device, the architecture was HATCN (10 nm)/TAPC (50 nm)/TCTa (10 nm)/mCBP: Firpic (10 wt%, 10 nm)/26DCzPPy: Firpic (10 wt%, 10 nm)/B3PYMPM (50 nm)/Liq (1.5 nm)/Al (130 nm). For the yellow-emitting device, the construction was HATCN (10 nm)/TAPC (50 nm)/TCTa (10 nm)/TCTa: PO-01 (5 wt%, 10 nm)/26DCzPPy: PO-01 (5 wt%, 10 nm)/B3PYMPM

(50 nm)/LiQ (1.5 nm)/Al (130 nm). For the red-emitting device, the configuration was HATCN (10 nm)/TAPC (50 nm)/TCTa (10 nm)/CBP: Ir(piq)₃ (5 wt%, 10 nm)/26DCzPPy: Ir(piq)₃ (5 wt%, 10 nm)/B3PYMPM (50 nm)/LiQ (1.5 nm)/Al (130 nm). (IV) NOA68 was spin-coated onto the Al cathode at 4000 rpm for 30 s, followed by UV curing under a 300 W lamp for 140 s. (V) The ADL (3M-VHB 4918) was mechanically laminated onto the surface of the NOA68, followed by peeling off the PDMS, thereby yielding ns-OLEDs with a top-emission configuration. In this process, the top-emission structure formed after stripping guarantees that the transmittance of the bottom ADL has no influence on the emission of the device, which occurs through the top IBL layer.

Characterization

The EL spectrum and current–voltage–luminance characteristics of the ns-OLEDs were measured using a Keithley 2450 digital source meter and a PR655 Luminance Meter from Photo Research. To measure the brightness of the ns-OLEDs under different strains, the device was first fixed on a Mark-10 mechanical testing platform with its two electrodes connected to a Keithley 2450 source meter. The luminance at various tensile strains was then measured using a PR-655 luminance meter. Optical microscopy images were acquired using an Olympus Microscope (BX51M, Olympus). 3D optical images were obtained by a 3D digital microscope (VHX-5000, Keyence). FIB-SEM images were measured using a focused ion beam scanning electron microscope (FIB-SEM, Helios NanoLab 660, FEI). AFM images were carried out on a Dimension Icon instrument (AFM, Dimension, Bruker). The Young's modulus of the samples was accurately measured in the mechanical mode of AFM by precisely controlling the peak force of the probe and analyzing the corresponding force–distance curves. X-ray photoelectron spectroscopy (XPS, ESCALB MKII, VG Scientific) was employed to analyze the internal functional groups and chemical bonding states of the materials. A four-probe system (RTS-9) was used to measure the sheet resistance of the electrode. The crack area ratio of the device after stretching was analyzed using ImageJ software.

Simulation

Finite-element software was performed using COMSOL Multiphysics 6.4 to model the multilayer structure subjected to external uniaxial strain and map the stress level. A three-dimensional model comprising functional layers (polymer anode, active layer, and metal cathode), two IBLs, and an ADL was constructed to analyze stress concentrations that may lead to a crack channeling across the thin film. The Young's modulus and Poisson's ratio of each layer were set to 33.6 GPa and 0.33 for the functional layers, 127.4 MPa and 0.4 for the IBL, and 15.2 kPa and 0.49 for the ADL, respectively. Meanwhile, the simulation was conducted for devices exposed to different UV illumination durations. The Young's modulus of the IBL was set to 15.6 MPa, 127.4 MPa, and 311.5 MPa for irradiation times of 20 s, 140 s, and 220 s, respectively. To simplify the simulation, we assume the adhesion between the layers is infinite, and no extra delamination occurs in the multilayer structure under uniaxial tensile strain up to 80%.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data supporting this study and its findings are available within the article and its Supplementary Information. Source data are provided with this paper.

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Author contributions

C.X., X.Z., and Q.T. conceived the original concept. C.X. designed and implemented the experiments, analyzed the data, and wrote the manuscript. X.Z. and Q.T. contributed to guiding the experiments, analyzing the data, and polishing the language. N.H. fabricated transparent anodes. Y.N., L.Y., J.Z., P.X., B.S., and M.Z. conducted AFM, FIB-SEM, and photographic characterization. X.Z., Q.T., Y.T., and Y.L. supervised the project. All authors contributed to the discussion of the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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